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Questions

Q1.

Which statement is **not** correct?

(1)

- ☒ **A** chlorine is more electronegative than bromine
- ☒ **B** chlorine is more reactive than bromine
- ☒ **C** chloride ions are stronger reducing agents than bromide ions
- ☒ **D** chloride ions are stronger reducing agents than fluoride ions

(Total for question = 1 mark)

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Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these is a disproportionation reaction?

- ☒ **A** $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$
- ☒ **B** $6NaOH + 3Br_2 \rightarrow 5NaBr + NaBrO_3 + 3H_2O$
- ☒ **C** $2NaOH + 2Al + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$
- ☒ **D** $NaOH + CO_2 \rightarrow NaHCO_3$

(Total for question = 1 mark)

Q3.

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A white solid **X** produces a red colour in a flame test.

When aqueous silver nitrate and nitric acid are added to a solution of **X**, a cream precipitate is formed which dissolves in concentrated aqueous ammonia.

What is the formula of **X**?

- ☒ **A** $SrBr_2$
- ☒ **B** NaBr
- ☒ **C** LiCl
- ☒ **D** BaI_2

(Total for question = 1 mark)

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Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Potassium chloride reacts with concentrated sulfuric acid, producing misty fumes.
It can be deduced that

- ☐ **A** sulfuric acid is acting as an oxidising agent
- ☐ **B** chloride ions are acting as an oxidising agent
- ☐ **C** hydrogen chloride is formed in the reaction
- ☐ **D** chlorine is formed in the reaction

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(Total for question = 1 mark)

Q5.

Which of these products are formed when chlorine is passed through cold, dilute aqueous sodium hydroxide?

(1)

- ☐ **A** NaCl and NaClO
- ☐ **B** NaClO and $NaClO_3$
- ☐ **C** NaCl and $NaClO_3$
- ☐ **D** NaClO and $NaClO_4$

(Total for question = 1 mark)

Q6.

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Potassium iodide reacts with concentrated sulfuric acid.

Which of the following is **not** a product?

(1)

- ☐ **A** H_2S
- ☐ **B** I_2
- ☐ **C** S
- ☐ **D** SO_3

(Total for question = 1 mark)

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Q7.

(a) Silver ions have anti-microbial properties and are used in some wound dressings. Silver nitrate can be made by warming a mixture of silver metal and concentrated nitric acid.

$$Ag(s) + 2HNO_3(aq) \rightarrow AgNO_3(aq) + NO_2(g) + H_2O(l)$$

Show, by reference to oxidation numbers, that this is a redox reaction.

(2)

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(b) Two students used different methods to determine the concentration of a silver nitrate solution.

(i) Student A used a calorimetric method, reacting a 50.0 cm^3 sample of the solution with excess powdered zinc.

$$AgNO_3(aq) + \tfrac{1}{2}Zn(s) \rightarrow \tfrac{1}{2}Zn(NO_3)_2(aq) + Ag(s) \quad \Delta_r H = -36.1\,kJ\,mol^{-1}$$

The student recorded a maximum rise in temperature of 5.2 °C.
Calculate the concentration of the silver nitrate solution in mol dm^{-3}.
[Assume the specific heat capacity of the solution is 4.18 J g^{-1} $°C^{-1}$ and the density of the solution is 1.00 g cm^{-3}.]

(3)

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(ii) Student B used a gravimetric method, which involved weighing a product of a reaction.
A 50.0 cm^3 sample of the same silver nitrate solution was mixed with excess potassium bromide solution.
The precipitate was filtered and weighed.

$$AgNO_3(aq) + KBr(aq) \rightarrow AgBr(s) + KNO_3(aq)$$

The mass of the precipitate was found to be 5.96 g.
Calculate the concentration of the silver nitrate solution, in mol dm^{-3}, from this gravimetric method.

(2)

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(iii) The students' values were different from the data book value.
Student A's value was lower and student B's value was higher.
Give a possible reason for each difference.

(2)

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(Total for question = 9 marks)

Q8.

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The products of the reaction of sodium fluoride with concentrated sulfuric acid can be predicted by considering the trends for the other sodium halides.

Which gas or gases form when sodium fluoride reacts with concentrated sulfuric acid?

- ☒ **A** hydrogen fluoride only
- ☒ **B** hydrogen fluoride and fluorine only
- ☒ **C** hydrogen fluoride and sulfur dioxide only
- ☒ **D** hydrogen fluoride, fluorine and sulfur dioxide only

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(Total for question = 1 mark)

Q9.

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This question is about the forces between molecules and ions.

*(a) Some data for three small molecules are shown.

Molecule	M_r	Boiling temperature / °C
Fluorine	38.0	–188
Hydrogen chloride	36.5	–85
Methanol	32.0	65

Explain the large variation in boiling temperatures, given the small range in M_r values. Detailed descriptions of the forces involved are not required.

(6)

(b) Calcium chloride is soluble in water.

Complete the diagram to show how water molecules interact with each ion.

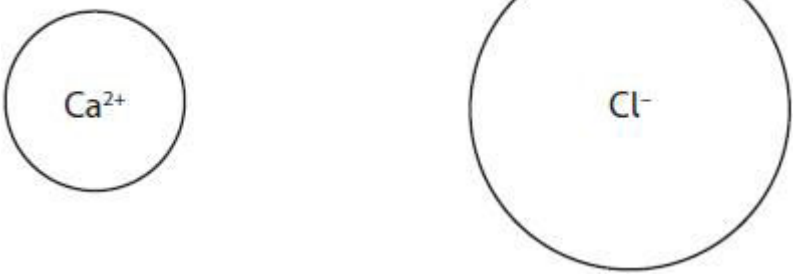
You may use  to represent a water molecule.

(2)

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(c) Explain why bromine is a liquid but iodine is a solid at room temperature.
Detailed explanations of the forces involved are not required.

(2)

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(Total for question = 10 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement about the Group 7 elements chlorine, bromine and iodine is **not** correct?

- ☒ **A** boiling temperature increases down the group
- ☒ **B** reactivity increases down the group
- ☒ **C** first ionisation energy decreases down the group
- ☒ **D** electronegativity decreases down the group

(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Solid potassium bromide reacts with concentrated sulfuric acid.

Which of these substances does **not** form?

- ☐ **A** bromine
- ☐ **B** hydrogen bromide
- ☐ **C** hydrogen sulfide
- ☐ **D** sulfur dioxide

(Total for question = 1 mark)

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Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When iodine is dissolved in a non-polar organic solvent, the solution formed is

- ☐ **A** purple
- ☐ **B** orange
- ☐ **C** colourless
- ☐ **D** brown

(Total for question = 1 mark)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which silver halides are soluble in **concentrated** aqueous ammonia?

- ☐ **A** AgBr and AgI
- ☐ **B** AgCl and AgI
- ☐ **C** AgCl and AgBr
- ☐ **D** AgCl only

(Total for question = 1 mark)

Q14.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these **increases** as Group 7 is descended?

☒ **A** oxidising ability of the molecular halogens

☒ **B** reducing ability of the halide ions

☒ **C** electrostatic attraction between the nucleus and outer shell of electrons

☒ **D** electronegativity of the halogen atoms

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(Total for question = 1 mark)

Q15.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which sequence shows the molecules in order of **increasing** boiling temperature?

☒ **A** $H_2O < Br_2 < Cl_2 < CH_4$

☒ **B** $Br_2 < CH_4 < Cl_2 < H_2O$

☒ **C** $Cl_2 < CH_4 < H_2O < Br_2$

☒ **D** $CH_4 < Cl_2 < Br_2 < H_2O$

(Total for question = 1 mark)

Q16.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Aqueous chlorine is added to an aqueous solution of potassium iodide.
A non-polar organic solvent is then added and the mixture is shaken.
The layers are allowed to separate.

What colour is seen in the organic layer?

☒ **A** brown

☒ **B** green

☒ **C** orange

☒ **D** violet

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(Total for question = 1 mark)

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Q17.

This question is about the elements in Group 7.

(a) Use your knowledge of the trends in the properties of Group 7 elements to predict the colour and physical state of astatine at room temperature.

(1)

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(b) (i) State the meaning of the term electronegativity.

(1)

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(ii) Explain the trend in electronegativity down Group 7.

(2)

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*(c) Compare and contrast the reactions of chlorine with

- water
- cold, dilute aqueous alkali
- hot, concentrated aqueous alkali

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Include an equation for each reaction, stating the type of reaction and the oxidation numbers of the chlorine involved. State symbols are not required.

(6)

(Total for question = 10 marks)

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Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reaction is **not** a disproportionation?

- ☐ **A** $3Cl_2(g) + 6NaOH(aq) \rightarrow NaClO_3(aq) + 5NaCl(aq) + 3H_2O(l)$
- ☐ **B** $2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$
- ☐ **C** $Cl_2(g) + H_2O(l) \rightarrow HCl(aq) + HClO(aq)$
- ☐ **D** $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$

(Total for question = 1 mark)

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Q19.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Four tests used to identify ions are shown:

1 flame test

2 addition of acidified barium nitrate solution

3 addition of acidified silver nitrate solution

4 addition of sodium hydroxide solution, then testing any gas with indicator paper

Which tests could be used to positively identify the ions in ammonium chloride?

- ☐ **A** 1 and 2
- ☐ **B** 1 and 3
- ☐ **C** 2 and 4
- ☐ **D** 3 and 4

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(Total for question = 1 mark)

Q20.

This question is about the halogens and some of their compounds.

(a) Descending the group from fluorine to iodine, the electronegativity of the atoms decreases even though their nuclear charge increases.

Explain the trend in electronegativity.

(2)

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(b) Iodate(V) ions, IO_3^-, react with iodide ions in acid solution.

$IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

Explain, in terms of the oxidation numbers of iodine in the three species, why this is **not** a disproportionation reaction.

(2)

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(c) Hydrogen bromide and hydrogen iodide reduce sulfuric acid.

Identify, by name or formula, the **compound** produced containing sulfur with its lowest oxidation number in that reaction.

(1)

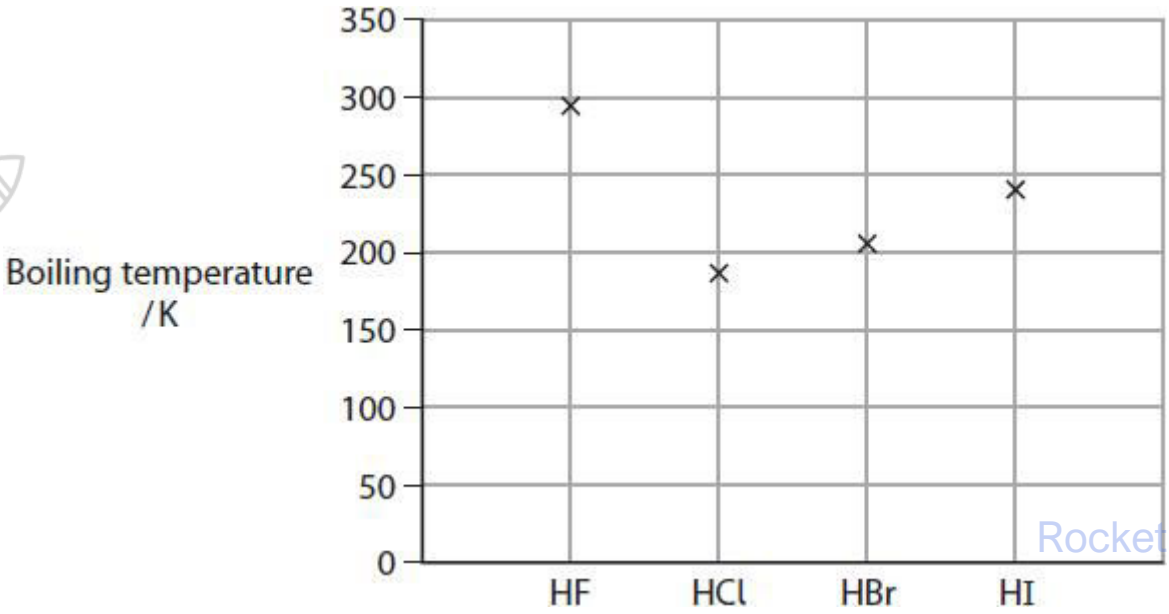
Hydrogen halide	Compound produced with the lowest oxidation number of sulfur
HBr	
HI	

(d) The graph shows the boiling temperatures of the hydrogen halides.

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Explain the trend in the boiling temperatures of the hydrogen halides.

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(e) A sample of seawater was evaporated to dryness.

The solid residue was weighed, then dissolved completely in deionised water.
Excess aqueous silver nitrate was added to the solution.
All the chloride ions in the seawater formed a precipitate of silver chloride.

$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

The precipitate was filtered, washed, dried and weighed.
Results:
Mass of solid residue from seawater = 0.098 g
Mass of silver chloride precipitate = 0.226 g
Calculate the percentage, by mass, of chloride ions in the solid residue.
[Assume silver chloride is the only precipitate.]

(3)

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(Total for question = 12 marks)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which trend is **not** correct as Group 7 is descended?

- ☒ **A** atomic radius of the elements increases
- ☒ **B** boiling temperature of the elements increases
- ☒ **C** electronegativity of the elements decreases
- ☒ **D** reactivity of the elements increases

(Total for question = 1 mark)

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Q22.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

When sodium bromide reacts with concentrated sulfuric acid, sodium hydrogensulfate is always formed.

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What other product(s) are formed in this reaction?

- ☒ **A** hydrogen bromide only
- ☒ **B** bromine and hydrogen bromide only
- ☒ **C** bromine, hydrogen bromide and sulfur dioxide only
- ☒ **D** bromine, hydrogen bromide, sulfur dioxide and sulfur only

(Total for question = 1 mark)

Q23.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A fixed amount of concentrated H_2SO_4 is reacted separately with an excess of four solid potassium halides.

In which reaction would the greatest number of moles of halide be oxidised?

- ☒ **A** $2KF(s) + H_2SO_4(aq) \rightarrow K_2SO_4(aq) + 2HF(g)$
- ☒ **B** $KCl(s) + H_2SO_4(aq) \rightarrow KHSO_4(aq) + HCl(g)$
- ☒ **C** $2KBr(s) + 3H_2SO_4(aq) \rightarrow 2KHSO_4(aq) + Br_2(l) + SO_2(g) + 2H_2O(l)$
- ☒ **D** $8KI(s) + 9H_2SO_4(aq) \rightarrow 8KHSO_4(aq) + 4I_2(s) + H_2S(g) + 4H_2O(l)$

(Total for question = 1 mark)

Q24.

This question is about some of the elements in Group 7 of the Periodic Table.

(a) Mixtures of halide salts are found in brine solutions extracted from oil and gas wells.
Iodine, which is used as a dietary supplement, may be obtained from these mixtures.

A brine solution containing 2.49 g of a mixture of potassium iodide and potassium chloride was analysed.

Procedure

Step 1 Excess aqueous silver nitrate solution was added to the solution to completely precipitate the halide ions.

Step 2 Excess aqueous ammonia was added to the mixture.

Step 3 The mixture was filtered, and the solid was washed, dried and weighed.

The mass of the dried solid was 0.162 g.

(i) State the colour of the solid in Step **3**.

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(1)

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(ii) Calculate the percentage by mass of potassium iodide in the mixture.
Give your answer to an appropriate number of significant figures.
[A_r values: Ag = 107.9 K = 39.1 I = 126.9]

(3)

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(b) Chlorine gas may be prepared by heating concentrated hydrochloric acid with solid manganese(IV) oxide.

$$MnO_2(s) + 4HCl(aq) \rightarrow MnCl_2(aq) + Cl_2(g) + 2H_2O(l)$$

Show, by reference to oxidation numbers, that this is a redox reaction.

(2)

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(c) Iodine may be obtained by bubbling chlorine gas through aqueous potassium iodide.

When the reaction is complete, hexane is added and the mixture shaken. Two layers are formed.
State the colour of each layer.

(2)

Aqueous layer

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Hexane layer

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* (d) Explain why iodine is more soluble in hexane than in water, by considering the intermolecular forces in iodine, hexane and water and any intermolecular forces formed between iodine and the solvents.

Detailed descriptions of how the intermolecular forces form are **not** required.

(6)

(Total for question = 14 marks)

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Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation shows a redox reaction that would **not** be expected to occur, based on the trend in reactivity of the halogens?

☐ **A** $I_2(aq) + 2At^-(aq) \rightarrow At_2(aq) + 2I^-(aq)$

☐ **B** $2Br^-(aq) + Cl_2(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$

☐ **C** $Cl_2(aq) + 2I^-(aq) \rightarrow I_2(aq) + 2Cl^-(aq)$

☐ **D** $2F^-(aq) + At_2(aq) \rightarrow 2At^-(aq) + F_2(aq)$

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(Total for question = 1 mark)

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Q26.

What are the gaseous products formed, other than water vapour, when concentrated sulfuric acid is added to potassium bromide?

(1)

☐ **A** bromine and sulfur dioxide only

☐ **B** bromine, hydrogen bromide and hydrogen sulfide only

☐ **C** bromine, hydrogen bromide and sulfur dioxide only

☐ **D** bromine, hydrogen bromide, sulfur dioxide and hydrogen sulfide only

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(Total for question = 1 mark)

Q27.

When chlorine is reacted with hot concentrated potassium hydroxide, the chlorine undergoes disproportionation.

What are the oxidation states of chlorine in the products?

(1)

☐ **A** –1 and +3

☐ **B** –1 and +5

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☒ **C** +1 and –1
☒ **D** +1 and +5

(Total for question = 1 mark)

Q28.

Which test is used to show that sodium chloride solution contains chloride ions?

(1)

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☒ **A** damp blue litmus paper turns red
☒ **B** damp blue litmus paper is bleached
☒ **C** dilute hydrochloric acid followed by silver nitrate solution gives a white precipitate
☒ **D** dilute nitric acid followed by silver nitrate solution gives a white precipitate

(Total for question = 1 mark)

Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The boiling temperatures of the halogens increase down Group 7 because on descending the group the

☒ **A** bond polarities increase
☒ **B** covalent bond strengths increase
☒ **C** electronegativities increase
☒ **D** number of electrons in the molecule increase

(Total for question = 1 mark)

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Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Bromine and chlorine each have two stable isotopes, ^{79}Br, ^{81}Br, ^{35}Cl and ^{37}Cl.
Which statement is true?

☒ **A** the atomic radius of ^{81}Br is greater than that of ^{79}Br
☒ **B** the electronegativity of ^{35}Cl is greater than that of ^{37}Cl
☒ **C** the first ionisation energy of ^{37}Cl is greater than that of ^{79}Br

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☒ **D** the mass spectrum of CH_2BrCl has two molecular ion peaks

(Total for question = 1 mark)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which solution remains clear on addition to barium chloride solution?

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☒ **A** dilute sulfuric acid
☒ **B** silver nitrate solution
☒ **C** sodium nitrate solution
☒ **D** sodium sulfate solution

(Total for question = 1 mark)

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Q32.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which substance is formed when concentrated sulfuric acid reacts with solid potassium chloride?

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☒ **A** Cl_2
☒ **B** S
☒ **C** H_2S
☒ **D** $KHSO_4$

(Total for question = 1 mark)

Q33.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Two tests were performed on an unknown ionic compound.

Silver nitrate solution was added to an aqueous solution of the compound, forming a white precipitate.

A flame test carried out on the ionic compound gave a red flame.

The unknown compound could have been

- ☐ **A** calcium iodide
- ☐ **B** lithium bromide
- ☐ **C** potassium carbonate
- ☐ **D** strontium chloride

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(Total for question = 1 mark)

Q34.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

What is the colour of the solution formed when iodine dissolves in cyclohexane?

- ☐ **A** blue-black
- ☐ **B** grey
- ☐ **C** purple
- ☐ **D** yellow

(Total for question = 1 mark)

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Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which row shows the correct trends **down** the group for the elements of Group 7?

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	Boiling temperature	Electronegativity	Reactivity as oxidising agent
☒ A	increases	decreases	decreases
☒ B	decreases	increases	increases
☒ C	increases	decreases	increases
☒ D	decreases	increases	decreases

(Total for question = 1 mark)

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Q36.

Cyclohexane (density = 0.78 g cm^{-3}) and aqueous iodine (density > 1.0 g cm^{-3}) are shaken and left to stand.
What colours are the two layers that form?

	lower layer	upper layer
☒ A	brown	yellow
☒ B	yellow	purple
☒ C	yellow	brown
☒ D	purple	brown

(Total for question = 1 mark)

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Q37.

Which is a displacement reaction that results in a visible colour change in solution?

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☐ **A** $2NaCl + At_2 \rightarrow 2NaAt + Cl_2$

☐ **B** $2NaBr + I_2 \rightarrow 2NaI + Br_2$

☐ **C** $2NaI + Br_2 \rightarrow 2NaBr + I_2$

☐ **D** $2NaF + Cl_2 \rightarrow 2NaCl + F_2$

(Total for question = 1 mark)

Q38.

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This question is about the reactions of some elements in Group 7.

(a) When chlorine water is added to a solution of potassium bromide, a pale brown colour is seen.

Write an ionic equation for this reaction.
State symbols are not required.

(1)

(b) Chlorine is used to treat drinking water.

$$Cl_2(g) + H_2O(l) \rightleftharpoons HClO(aq) + HCl(aq)$$

By considering oxidation numbers, show that this equation represents a disproportionation reaction.

(2)

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(c) Concentrated sulfuric acid, H_2SO_4, reacts with solid potassium halides.

Some observations are shown in the table.

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Halide	Observations
Potassium chloride	Misty fumes
Potassium bromide	Misty fumes and brown vapour
Potassium iodide	Misty fumes and purple vapour

(i) The equation for the reaction of concentrated sulfuric acid with potassium chloride is shown.

$$H_2SO_4 + KCl \rightarrow HCl + KHSO_4$$

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State why this is **not** a redox reaction.

(1)

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(ii) The half-equation for the oxidation of iodide ions is shown.

$$2I^- \rightarrow I_2 + 2e^-$$

The half-equation for the reduction of concentrated sulfuric acid to hydrogen sulfide is shown.

$$8H^+ + H_2SO_4 + 8e^- \rightarrow H_2S + 4H_2O$$

Deduce the ionic equation for the reaction between iodide ions and concentrated sulfuric acid by combining these two half-equations.

(1)

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(iii) Explain the difference in the reaction of concentrated sulfuric acid with chloride ions and with iodide ions.

(2)

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(Total for question = 7 marks)

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Q39.

Concentrated sulfuric acid reacts with sodium iodide.

What is one role of the sulfuric acid?

- ☒ **A** oxidising agent
- ☒ **B** reducing agent
- ☒ **C** proton acceptor
- ☒ **D** lone pair donor

(Total for question = 1 mark)

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Q40.

Which equation is **not** an example of disproportionation?

☒ **A** $Cu_2O + H_2SO_4 \rightarrow Cu + CuSO_4 + H_2O$

☒ **B** $3HNO_2 \rightarrow HNO_3 + 2NO + H_2O$

☒ **C** $3ClO^- \rightarrow 2Cl^- + ClO_3^-$

☒ **D** $Mg + 2HCl \rightarrow MgCl_2 + H_2$

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(Total for question = 1 mark)

Q41.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An aqueous solution of potassium chloride is added to a solution of iodine in hexane and shaken. After a time, two layers are formed.

What is the colour of each layer?

	Upper layer	Lower layer
☒ **A**	purple	yellow
☒ **B**	yellow	purple
☒ **C**	purple	colourless
☒ **D**	colourless	purple

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(Total for question = 1 mark)

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